

WAA — Board Review 2026

Theoretical Optics: Consolidated Review

Personal study notes / Sources: MCU 2024 (Dr. Viñola & Dr. De Mesa), Board Primer, MCU Quizzes

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1. Optics: Definition and Branches

Optics is the branch of science that deals with the nature, behavior, and properties of light and its interaction with matter. The word comes from the Latin *optikes*.

Main Branches

Branch	What it studies
Geometrical Optics	Light as rays; image formation by lenses, mirrors, prisms
Physical / Wave Optics	Light as waves; interference, diffraction, polarization
Quantum / Photon Optics	Light as energy packets (quanta/photons)
Mechanical / Ophthalmic Optics	How the eye uses lenses, prisms, mirrors; lens dispensing
Visual / Physiological Optics	The eye as an optical system

I On the board exam: Geometrical Optics = light as **ray**. Physical Optics = light as **wave**.

2. Nature and Properties of Light

Light is a form of radiant energy emitted by luminous bodies at 186,000 miles/sec (300,000 km/sec). It is electromagnetic radiation — the visible portion of the EM spectrum that stimulates the retina.

Properties of Light

1. Form of radiant energy — can be transferred through space
2. Emitted by luminous bodies
3. Travels fastest in vacuum — the speed in vacuum is a universal constant
4. Only a small portion of the EM spectrum that can stimulate the retina
5. White light is polychromatic (made of multiple wavelengths)
6. Makes things visible and vision possible
7. Can be **reflected**, **refracted**, or **absorbed**

- 8. Can convey messages and information

What Happens When Light Hits an Object

- **Reflection** — bounces back at the surface
- **Refraction** — bends as it passes into a different medium
- **Absorption** — converted to heat or other energy

3. Speed of Light — Historical Milestones

Person	Year	Contribution
Galileo	~1600	First attempted measurement; concluded light travels at infinite speed
Ole Romer	1675	Proved speed is finite by observing Jupiter's moon (Io)
Armand Fizeau	1849	First to measure speed through land (Toothed Wheel Experiment)
Leon Foucault	1850	First to measure speed of light in water (Rotating Mirror Experiment)

Speed in vacuum: 186,000 miles/sec = 300,000 km/sec

Speed in a medium:

Speed in medium = 186,000 miles/sec / n

Where n = index of refraction of that medium.

Higher index = slower speed. Polycarbonate ($n=1.58$) slows light more than crown glass ($n=1.523$).

4. Theories of Light

Early (Pre-Modern) Theories

Theorist	Era	Claim
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Pythagoras (600 BC)	Ancient	Visible objects are sources of light
Empedocle (500 BC)	Ancient	The eye is the source of light
Plato (500 BC)	Ancient	Both the eye and the object are sources

Modern Theories

Theory	Alternate Name	Proponent	Year	Core Idea
Corpuscular Theory	Emission / Particle Theory	Isaac Newton	1642	Light = tiny particles (corpuscles) traveling in straight lines
Wave Theory	Undulatory Theory	Christian Huygens	1690	Light = wave motion spreading from source through elastic medium (ether)
Electromagnetic Theory	—	James Clerk Maxwell	1865	Light = electromagnetic wave; electricity and magnetism are unified
Quantum Theory	Photon Theory	Max Planck / Einstein	1900	Light = discrete energy packets (quanta/photons); atoms emit light in packets

Board exam tip: "Emission Theory" = corpuscles (Newton). "Corpuscular Theory" and "Particle Theory" are the same thing.

Thomas Young (1801) demonstrated interference, supporting the Wave Theory.

Feynman Cross-Check

Think of it this way: Newton saw light like a stream of bullets (straight lines, casts sharp shadows).

Huygens saw it like ripples on water (waves spreading out). Maxwell proved those ripples are electric and magnetic fields. Planck showed the "ripples" come in countable packets — that is the photon.

5. Sources of Light

By Origin

- **Natural sources** — cannot be controlled by humans (sun, stars, fire, bioluminescent organisms)
- **Artificial sources** — controlled by humans
 - *Incandescent* — light from heat (flames, arc lamps, glowing coals)
 - *Luminescent* — light from excited electrons dropping to lower energy (fluorescent lamps, gas discharge, LEDs)

By Size

- **Point source** — smallest possible light source; produces only umbra
- **Extended source** — has measurable size; produces umbra + penumbra

Luminous vs. Illuminated Objects

- **Luminous** — generates its own light
 - **Illuminated** — reflects light to our eyes
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6. Physical / Wave Optics

Types of Waves

Type	Description
Longitudinal	Particle displacement is parallel to wave motion (e.g., sound)
Transverse	Particle displacement is perpendicular to wave motion (light)

Transverse waves come in two types:

- **Travelling / Progressive** — moves outward from source to infinity
- **Standing / Stationary** — results from interference of reflected waves; has nodes (rest) and antinodes

(max displacement)

Characteristics of a Wave

Term	Definition
Crest	Upward peak of the wave
Trough	Downward valley of the wave
Amplitude	Maximum displacement from equilibrium = height of crest; determines brightness/intensity
Wavelength (λ)	Distance for one complete vibration; distance between two successive crests
Frequency (f)	Number of complete waves per second (unit: Hertz, Hz)
Period (T)	Time for one complete cycle
Velocity (v)	Speed of wave travel

Wave Equation:

$$\text{Velocity} = \text{Wavelength} \times \text{Frequency}$$
$$v = \lambda \times f$$

When light passes from one medium to another: **velocity and wavelength change, frequency stays the same.**

Units of Wavelength

- Angstrom ($\text{\AA}$) — 1/10,000,000,000 of a meter (10^{-10} m)
- Micron / Micrometer (μm) — 1/1,000,000 of a meter (10^{-6} m)
- Nanometer (nm) / Millimicron — 1/1,000,000,000 of a meter (10^{-9} m)

Wave Interference

When two waves meet in the same medium:

Type	What happens	Phase
Constructive	Crest + crest = super-crest; trough + trough = super-trough	Waves in phase
Destructive	Crest + trough = cancellation	Waves out of phase

Superposition Principle (Thomas Young, 1801): When two waves interfere, the resulting displacement at the crossing point is the algebraic sum of the individual displacements.

Examples of interference in real life:

- Colors in soap bubbles
 - Brilliant colors of opal
 - Colors inside abalone shells
 - Colors in oil films on water
 - Anti-reflective coatings on lenses
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Diffraction

The apparent bending of waves around small obstacles and spreading through small openings.

- If the opening is much larger than the wavelength — bending is minimal
- If the opening is about the same size as the wavelength — bending is considerable

Diffraction grating — optical component that splits (diffracts) light into beams traveling in different directions; acts as a dispersive element.

Polarization

The process of transforming unpolarized light into polarized light.

- **Unpolarized light** — vibrates in multiple planes (e.g., sunlight, lamp, candle)
- **Polarized light** — vibrates in a single plane

Polaroid filters block one plane of vibration, reducing glare and light intensity by half.

Benefits of polarized lenses in optometry:

- Reduces eyestrain and reflections
 - Improves visual comfort and clarity
 - Enhances color and depth perception
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7. EM Spectrum and Radiation Effects on the Eye

EM Spectrum (shortest to longest wavelength)

Gamma rays → X-rays → Ultraviolet → **Visible** → Infrared → Microwaves → Radio waves

Visible Spectrum

- Range: approximately 380–760 nm
- Red = longest wavelength, lowest frequency
- Violet = shortest wavelength, highest frequency

Invisible Radiation and the Eye

Radiation	Wavelength	Ocular Effect
Short UV (below 380 nm)	Shortest	Can cause photokeratitis, cataract
Short infrared (750–1,500 nm)	Just above visible	Relatively safe in small doses
Long infrared (1,500 nm and up)	Long IR	Absorbed by the cornea; prolonged exposure causes conjunctivitis; sources: direct sunlight, molten glass/metal, infrared lamps
X-rays	Very short	Medical imaging; ionizing
Gamma rays	Smallest wavelength / highest frequency	Cancer treatment; ionizing

Draper's Law: For radiation to have an effect on a substance, it must first be **absorbed** by that substance.

Effects of absorbed radiation:

- **Thermal effect** — heating due to absorption (infrared)
 - **Photochemical effect** — chemical reactions triggered by light absorption (UV, visible)
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8. Subdivisions of Light

1. Light Ray

The path of a single photon from a single point on a light source.

2. Pencil of Light (Bundle of Rays)

Type	Description
Divergent pencil	Rays spread apart from a point source; comes from a near source
Convergent pencil	Rays come together toward a single point
Parallel pencil	Rays never meet; from a source at optical infinity

3. Beam of Light

A collection of divergent, convergent, or parallel pencils arising from an **extended source**.

 A single point source produces pencils. An extended source produces beams.

9. Vergence

Vergence expresses how much light is converging or diverging at a specific point in its path. Think of it as the "power" the light has to converge or spread at that location.

Formula

$$V = 1 / d$$

- **V** = vergence (diopters, D)

- **d** = distance from the point in question to the source or focus (in meters)
- Use the sign convention rule below

Sign Convention Rule

1. Assume light travels left to right
2. Distance measured in the same direction light travels → positive (+)
3. Distance measured opposite to light travel → negative (-)

Vergence by Pencil Type

Pencil	Vergence
Convergent	Positive (+)
Divergent	Negative (-)
Parallel	Zero (0)

Formula depth: The reciprocal relationship ($1/d$) tells you how "urgently" the light wants to meet a point. At 1 meter, $V = 1$ D. At 0.25 m, $V = 4$ D — much more urgent. This is the same concept as lens power: a +4.00 D lens brings parallel rays to a focus at 25 cm. Vergence at 25 cm from a point source is also 4 D. Cross-subject connection: accommodative demand equals vergence of the near object; a target at 40 cm requires 2.5 D of accommodation.

10. Rectilinear Propagation

"In a homogeneous and isotropic medium, light travels in a straight line."

"Homogeneous" means the medium is uniform throughout. "Isotropic" means light behaves the same in every direction within it. The moment either condition changes (different density, a boundary), light can bend or bounce.

This is the law behind:

- Shadow formation (light cannot bend around objects)
- The pinhole camera (sharp images through a tiny hole)
- Eclipses (the moon blocks the sun's rays in a straight path)

11. Shadow Formation

A shadow is a dark image formed when an opaque object intercepts light rays.

With a Point Source

- Only **umbra** is formed (total/absolute/geometric shadow)
- Sharp boundary
- The shadow is the cone formed by straight rays tangent to the object's surface, with the apex at the source

Size rules:

- Move the source closer to the object → shadow gets bigger
- Move the object closer to the screen → shadow gets smaller

With an Extended Source

Two shadow regions form:

Region	Description
Umbra	Central, totally dark region with sharp border (Latin: "shade")
Penumbra	Outer annular region, partly shaded (Latin: "paene" = nearly)

Shadow Calculations (Ratio and Proportion)

For a point source (single shadow = umbra):

$$a / b = c / d$$

- a = object (height or diameter)
- b = distance from source to object
- c = shadow size
- d = total distance from source to screen

For an extended source (umbra and penumbra):

$$a_1 / b = x / (b + d) \quad \leftarrow \text{for first partial triangle}$$

$$a_2 / b = y / (b + e) \quad \leftarrow \text{where } e = \text{distance on other side}$$

Umbra diameter = $x - y$

Penumbra diameter = $x + y$

Area = πr^2

High-Yield Examples

Example 1 (point source):

Object diameter = 8 cm, 35 cm from source; screen 75 cm beyond the object (total = 110 cm from source).

Shadow size:

$$8/35 = c/110 \rightarrow c = 880/35 = 25.14 \text{ cm}$$

Example 2 (shadow of aperture, point source):

Point source 15 cm from aperture (diameter 5 cm); screen 45 cm beyond aperture. Diameter of shadow:

$$a/b = c/d \rightarrow 5/15 = c/(15+45) \rightarrow c = 5 \times 60/15 = 20 \text{ cm}$$

Region of Illumination (Extended Source Through Aperture)

- **Full illumination** — uniform central region
- **Partial illumination** — outer annular region with partial darkness

12. Eclipses

Eclipses demonstrate the Law of Rectilinear Propagation of Light in nature.

Eclipse	Arrangement	What is the object
Solar Eclipse	Sun → Moon → Earth (SME)	Moon blocks sun; Earth enters moon's shadow cone
Lunar Eclipse	Sun → Earth → Moon (SEM)	Earth blocks sun; Moon enters Earth's shadow cone

Memory: SME = Solar (Moon in the middle). SEM = Lunar (Earth in the middle).

Apparent sizes of sun and moon:

- The sun is about 400× larger than the moon, but also about 400× farther away — so they appear the same angular size, making total solar eclipses possible.
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13. Pinhole Camera

A lensless device consisting of a light-tight box with a small hole and dark interior. Also called *Camera Obscura* (darkened room) or Porta's Camera.

- **Invented by:** Alhazen
- **Perfected by:** Giambattista de la Porta

Properties of Pinhole Camera Image

1. Image is **real and inverted**
2. Clarity depends on the size of the hole
3. Optimum hole diameter is proportional to the camera's length
4. Image size depends on: camera length and distance of object from hole
5. **Large depth of focus** (almost everything in focus)
6. Low illumination compared to lens cameras
7. Image is free from distortion

Abney's Formula (Optimum Pinhole Size)

$$y = k \times x$$

- **y** = pinhole diameter
- **x** = length of camera
- **k** = constant:
 - $k = 0.008$ when using inches
 - $k = 0.01275$ when using centimeters

Ratio-Proportion Formula for Image Size

$$a / b = a' / b'$$

- a = altitude/height of smaller triangle (object)
 - b = base of smaller triangle (distance: source to hole)
 - a' = altitude of bigger triangle (image or shadow)
 - b' = base of bigger triangle (total distance from source)
-

14. Angular Elevation

The angle of elevation is the angle between a horizontal line and the line of sight up to an object.

Formula

$$\tan \theta = h / d$$

Rearranging to find the distance:

$$d = h / \tan \theta$$

Example: A building is 5 ft tall. The angle of elevation to its top is 20° . How far is the observer from the base?

$$d = 5 / \tan 20^\circ = 5 / 0.364 = 13.74 \text{ ft}$$

15. Diopter and Focal Length

The **diopter** is the unit of lens power — or the vergence of light at a specific point. It is the reciprocal of the focal length in meters.

Diopter Formulas

Unit of focal length	Formula
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Meters (m)	$D = 1/f$
Centimeters (cm)	$D = 100/f$
Millimeters (mm)	$D = 1000/f$
Inches (in)	$D = 40/f$

Formula depth: The "40" comes from the conversion 1 inch = 2.54 cm, so 1 meter = 39.37 $\approx$ 40 inches. The constant is a convenience so you can plug inches directly without converting.

Cross-subject connections:

1. These are the same formulas used to calculate accommodative amplitude from near-point measurements
2. Lensometer readings are also in diopters (same unit)

High-Yield Worked Examples

1. Focal length $\rightarrow$ power:

A diverging lens has $f = 30.77$ cm. Power = $100/30.77 = -3.25$ D (negative because diverging)

2. Power $\rightarrow$ focal length in inches:

Power = -2.75 D. $f = 40/2.75 = 14.55$ inches

3. Focal length 15 inches $\rightarrow$ power:

$D = 40/15 = +2.67$ D

4. Diverging lens, 8 inches focal length $\rightarrow$ power:

$D = 40/8 = -5.00$ D

16. Optical Mediums

Medium	Definition	Example
Transparent	Light passes freely through; minimum absorption and reflection	Clear glass, water

Opaque	All incident rays are absorbed or reflected; no transmission	Wood, metal
Translucent	Partially transmits light; objects not clearly visible	Frosted glass, tissue paper

17. Index of Refraction

The index of refraction measures how much light slows down (and bends) when it enters a material compared to its speed in vacuum.

$$n = \text{speed of light in vacuum} / \text{speed of light in medium}$$

Higher index = slower speed in the medium = more bending.

Index of Refraction Table (Board-Relevant Values)

Material	n
Vacuum / Air	1.000
Water	1.333
Crown glass (ophthalmic standard)	1.523
Polycarbonate	1.586
Light flint glass	1.58
Dense flint glass	1.65
Diamond	2.417
Cornea	1.376
Aqueous humor	1.336
Crystalline lens cortex	1.386
Crystalline lens nucleus	1.406

Vitreous humor	1.336
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Abbe Value and Chromatic Aberration

Index	Thickness	Abbe value	Chromatic aberration
Higher	Thinner lens	Lower	Higher
Lower	Thicker lens	Higher	Lower

The higher the index → the lower the Abbe value → the higher the chromatic aberration.

18. Dispersion and Chromatic Aberration

Dispersion is the decomposition of white light into its component wavelengths when it passes through a medium with non-parallel surfaces (such as a prism).

White light separates because different wavelengths travel at slightly different speeds through a medium, so each bends at a slightly different angle.

Color	Wavelength	Frequency	Bending
Red	Longest	Lowest	Least
Orange	—	—	—
Yellow	—	—	—
Green	—	—	—
Blue	—	—	—
Indigo	—	—	—
Violet	Shortest	Highest	Most

- Shorter than violet = **ultraviolet** (UV)

- Longer than red = **infrared** (IR)

Chromatic aberration in ophthalmic lenses occurs because a lens also disperses light — different wavelengths focus at slightly different points. Higher-index materials have more chromatic aberration (lower Abbe value).

19. Reflection

Reflection is the bouncing back of light at the surface of a medium.

Factors Affecting Reflection

- Physical properties of the reflecting surface
- Type of surface (smooth vs. rough)
- Angle of incidence
- Wavelength of light

Laws of Reflection

1. The angle of incidence equals the angle of reflection (both measured from the normal)
2. The incident ray, the normal, and the reflected ray all lie in the same plane

$$i = r$$

Types of Reflection

- **Regular (specular) reflection** — smooth surfaces (glass, mirror, polished metal); rays reflect in one direction
- **Irregular (diffuse) reflection** — rough surfaces; light scatters in multiple directions

Images

- **Real image** — formed by actual intersection of reflected/refracted rays; can be projected on a screen
- **Virtual image** — formed by the prolongation (extension) of rays; cannot be projected
- **Pervverted image** — erect but laterally reversed (what you see in a plane mirror)

Multiple Images in Plane Mirrors

$$\text{Number of images} = (360 / a) - 1$$

Where **a** = angle between the two mirrors (in degrees)

Alternate formula: $2(180)/a - 1$

Example: Mirrors at $60^\circ \rightarrow \text{Images} = 360/60 - 1 = \mathbf{5 \text{ images}}$

Deviation Produced by a Plane Mirror

$\text{Deviation} = 180^\circ - 2(\text{angle of incidence})$

Example: $i = 30^\circ \rightarrow \text{Deviation} = 180 - 60 = \mathbf{120^\circ}$

20. Spherical Mirrors

A spherical mirror is a section of a sphere — concave (curves inward) or convex (curves outward).

Cardinal Points

Point	Description
Center of Curvature (C)	Center of the sphere; equidistant to all points on the mirror
Principal Focus (F)	Where parallel, near-axis rays converge (concave) or appear to diverge from (convex)
Vertex (V)	Center of the mirror's aperture; where the principal axis meets the mirror
Radius of Curvature (R)	Distance from vertex to center of curvature; $R = 2f$

Three Behaviors of Light in Ray Tracing

1. Ray parallel to the principal axis $\rightarrow$ reflects through (or as if from) the principal focus
2. Ray through (or directed toward) the principal focus $\rightarrow$ reflects parallel to the principal axis
3. Ray through (or directed toward) the center of curvature $\rightarrow$ reflects back on its own path

Mirror Formula

$$1/f = 1/p + 1/q$$

- **f** = focal length
- **p** = object distance (always positive)
- **q** = image distance (positive if real, negative if virtual)

Magnification:

$$M = -q/p = \text{image height} / \text{object height}$$

- M is negative → real image (inverted)
- M is positive → virtual image (upright)
- $|M| > 1$ → magnified
- $|M| < 1$ → minified

Relationship between R and f:

$$f = R/2$$

Sign Conventions for Mirrors

Quantity	Concave	Convex
f	+	-
R	+	-
q	+ (real)	- (virtual)
M	- (real, inverted)	+ (virtual, upright)

Image Formation in Concave Mirror (6 Cases)

Object position	Image location	Nature	Orientation	Size
Beyond C (object > 2f)	Between F and C	Real	Inverted	Minified
At C (object = 2f)	At C	Real	Inverted	Same size
Between C and F (f < object < 2f)	Beyond C	Real	Inverted	Magnified
At F	No image	—	—	Image at infinity
Between F and V (object < f)	Behind mirror	Virtual	Upright	Magnified
Behind mirror	In front	(not standard)	—	—

Image Formation in Convex Mirror

Regardless of object position:

- Image is always **virtual, upright, and minified**
- Located behind the mirror (between V and F)

21. Refraction

Refraction is the bending of light as it passes from one medium to another of different optical density. The velocity changes, causing the direction to change.

Laws of Refraction (5 Laws)

1. A ray striking a refracting surface perpendicularly is **undeviated**
2. Going from less dense to more dense → ray bends **toward the normal**
3. Going from more dense to less dense → ray bends **away from the normal**
4. The incident ray, the normal, and the refracted ray all lie at the point of incidence
5. **Snell's Law** — the sine of the angle of incidence is to the sine of the angle of refraction as the inverse ratio of their indices

Snell's Law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- n_1 = index of first medium
- θ_1 = angle of incidence
- n_2 = index of second medium
- θ_2 = angle of refraction

Physical analogy: Think of running from the beach into water. Your feet slow down when they hit the water first, pulling you toward the perpendicular. That is refraction — the part of the wavefront entering the denser medium slows down first, bending the entire ray toward the normal.

Cross-subject connections:

1. The eye refracts light at the tear film, cornea, aqueous, lens, and vitreous — each interface follows Snell's Law
2. Index of refraction values determine the power of each refractive surface (lensmaker's equation)

Worked Example

Air ($n=1.00$) to water ($n=1.33$), angle of incidence = 40° :

$$\begin{aligned} 1.33 \times \sin 40^\circ &= 1.00 \times \sin R \\ 1.33 \times 0.6428 &= \sin R \\ \sin R &= 0.8549 \rightarrow R = 58.75^\circ \end{aligned}$$

Absolute vs. Relative Index

- **Absolute index** = ratio of speed of light in air/vacuum to speed in medium
- **Relative index** = ratio comparing two media (neither is air)

$$n_{\text{relative}} = v_1 / v_2$$

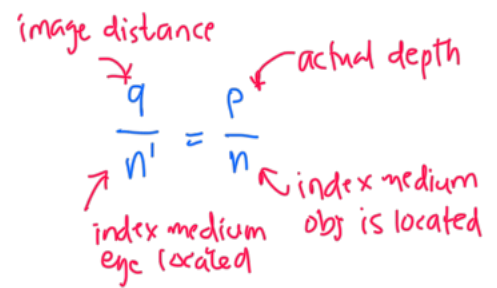
Lateral Displacement

When light passes through a **plane refractor** (two parallel surfaces, like a glass slab), it exits parallel to its original direction but displaced sideways. This sideways shift is the **lateral displacement**.

No net change in direction (comes out parallel), just a shift in position.

$$d = \frac{t \sin(i - R)}{\cos R}$$

↑ thickness, ↓ incidence, ↓ refracted, ↑ displacement, COS R, ← refracted



Apparent / Linear Displacement

When viewing an object in a different medium, it appears closer or farther than it really is.

$$\text{Apparent depth} = \text{Real depth} \times (n_{\text{observer}} / n_{\text{object}})$$

Example: A fish in water ($n=1.33$) at 20 cm real depth appears at: $20 \times (1.00/1.33) = \mathbf{15 \text{ cm}}$ from the observer.

22. Critical Angle and Total Internal Reflection

Critical Angle

The critical angle is the specific angle of incidence (in the denser medium) at which the refracted ray exits at exactly 90° to the normal — meaning it travels along the surface, not into the other medium.

Conditions:

- Light travels from denser to rarer medium ($n_1 > n_2$)
- Angle of refraction = 90°

Formula:

$$\sin \theta_c = n_2 / n_1$$

Material	Critical angle in air
Dense flint glass ($n=1.65$)	$\sin^{-1}(1/1.65) = \mathbf{37.31^\circ}$
Diamond ($n=2.417$)	$\sin^{-1}(1/2.417) = \mathbf{24.4^\circ}$
Crown glass ($n=1.523$)	$\sin^{-1}(1/1.523) = \mathbf{41.0^\circ}$

Total Internal Reflection (TIR)

When the angle of incidence exceeds the critical angle, no refraction occurs — all light bounces back into the dense medium. This is Total Internal Reflection.

Conditions for TIR:

1. Light is in the denser medium *high $n \rightarrow$ low n*
2. Angle of incidence $>$ critical angle *$\angle i > \angle c$*

Applications:

- Optical fibers (fiber optics in OCT, endoscopy)
 - Gonio prisms work because TIR is broken by applying gel of matching n
 - Diamond's brilliance comes from its low critical angle (much internal reflection)
 - Prisms in retinoscopes and ophthalmoscopes
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23. Lenses

Classification

Type	Shape	Effect on Light	Sign
Convex (convergent, plus)	Thicker at center	Converges light	(+)
Concave (divergent, minus)	Thinner at center	Diverges light	(-)

Physical Principles

A lens acts as a series of prisms: the thicker the lens (more prism-like the edges), the more it bends light. A convex lens bends light toward the base of each prism (toward the optical axis), converging rays. A concave lens bends light toward the apex, diverging rays.

Lens Nomenclature (Periscopic Lenses)

- Plus lenses: back surface = -1.25 D
- Minus lenses: front surface = $+1.25$ D

Cardinal Points of a Lens

- **Optic axis** — line joining the centers of curvature of both lens surfaces

- **Optic center** — point through which rays pass undeviated (no bending)
- **Principal bending plane** — perpendicular to the optic axis through the optic center; where object and image distances are measured

Thin Lens Formula

$$1/f = 1/p + 1/q$$

- **p** = object distance (positive, measured from lens)
- **q** = image distance (positive if real, on the opposite side of lens from object)
- **f** = focal length (positive for convex, negative for concave)

Magnification:

$$M = -q/p = \text{image height} / \text{object height}$$

Image Formation by Convex Lens (6 Cases)

Object position	Image	Nature	Orientation	Size
Beyond 2F	Between F and 2F	Real	Inverted	Minified
At 2F	At 2F	Real	Inverted	Same size
Between F and 2F	Beyond 2F	Real	Inverted	Magnified
At F	Infinity	No image	—	—
Between F and lens	Same side as object	Virtual	Upright	Magnified

Image Formation by Concave Lens

Regardless of object position:

- Image is always **virtual, upright, and minified**
- Located on the same side as the object

Worked Example

A tree 6 m high is 3 m from a concave lens ($f = 1 \text{ m}$). Nature of image?

$$1/f = 1/p + 1/q$$

$$1/(-1) = 1/3 + 1/q$$

$$1/q = -1 - 1/3 = -4/3$$

$$q = -0.75 \text{ m (virtual, same side as object)}$$

→ Virtual, erect, minified

Lensmaker's Equation (Surface Power)

$$F = (n' - 1)(1/r_1 - 1/r_2)$$

Where r_1 and r_2 are the radii of curvature of the front and back surfaces.

24. Cylinder Lenses

A cylinder lens has different curvatures in different meridians:

- **Power meridian** — meridian of maximum power (90° from the axis)
- **Axis meridian** — meridian of minimum (or zero) power

Astigmatism arises when there are two different curvatures on a single refracting surface or in the optical system — similar to what a cylinder lens deliberately creates.

Power in an Oblique Meridian

For a **simple cylinder**:

$$D_{\text{oblique}} = F_c \times \sin^2 \alpha$$

For a **sphero-cylinder**:

$$D_{\text{oblique}} = F_s + F_c \times \sin^2 \alpha$$

- **D_oblique** = power in the oblique meridian
- **F_s** = sphere power
- **F_c** = cylinder power

- α = angle between the cylinder and the oblique meridian
-

25. Thick Lens Formulas

Thin Lens Approximation

$$F_a = F_1 + F_2$$

Used when lens thickness is negligible.

Equivalent Power (True Power)

$$D_{eq} = F_1 + F_2 - (t/n) \times F_1 \times F_2$$

- t = lens center thickness (in meters)
- n = index of refraction of the lens material
- The (t/n) term accounts for the reduced distance between the two surfaces inside the lens

Formula depth: When light travels through a thick lens, it is refracted at the front surface, travels a distance t through the lens material (but effectively travels t/n in air-equivalent distance), then refracts again at the back surface. The product $F_1 \times F_2$ accounts for the interaction between the two surfaces. This is why thick plus lenses have slightly less power than predicted by the thin-lens formula.

Cross-subject connections:

1. Back vertex power (BVP) is what the lensometer measures — not equivalent power
2. Thick lens power matters for high-powered glasses (above ± 5.00 D) and contact lens back vertex calculations

Back Vertex Power (BVP) / Effective Power

$$D_{BV} = F_1 + F_2 + (t/n) \times F_1^2$$

Measured from the back surface. This is what the **lensometer** reads.

Front Vertex Power / Neutralizing Power

$$D_{FV} = F_1 + F_2 + (t/n) \times F_2^2$$

Measured from the front surface.

26. Oblique Power, Spherical Equivalent, Pantoscopic Tilt

Spherical Equivalent

$$SE = \text{Sphere} + \frac{1}{2} \times \text{Cylinder}$$

Used when a single spherical value must represent a sphero-cylindrical prescription.

Example: $-1.00 \text{ S} -1.50 \text{ C} \times 180 \rightarrow SE = -1.00 + (-0.75) = -1.75 \text{ D}$

Pantoscopic Tilt

When a lens is tilted forward (pantoscopic angle), the effective power and cylinder change:

New sphere after tilt:

$$S' = S_o / \cos^2 \alpha$$

Induced cylinder:

$$C = S' \times \tan^2 \alpha$$

- **S'** = new sphere power
 - **S_o** = original sphere power
 - **α** = angle of obliquity (tilt)
-

27. Prisms

What a Prism Does

A prism is a wedge-shaped transparent optical medium bounded by two non-parallel, polished surfaces.

- Light passing through a prism **deviates toward the base**
- An object viewed through a prism **appears displaced toward the apex**
- Prisms **do not change vergence** (surfaces are flat; no dioptric power)
- Prisms **do not form images** (no focal point)
- White light through a prism is **dispersed** into the spectrum

Optical Properties Summary

- Convergent rays entering a prism → still converge when leaving
- Divergent rays entering → still diverge when leaving (just shifted in direction)
- The deviation increases with prism power and index

Angle of Deviation Formula

$$\text{Dev} = a_A \times (n - 1)$$

- **a_A** = apical angle of the prism
- **n** = index of refraction of the prism material

Units of Prism Power

Unit	Definition	Displacement at 1 meter
Prism Diopter (Δ)	Displaces image 1 cm at 1 meter distance	1 cm
Prism Degree (°d)	Produces deviation of 1 degree	1.75 cm
Centrad (∇)	Displaces image 1/100th of a radian arc at 1 meter	0.5729 cm

Memory: Prism Diopter > Centrad > Prism Degree (in terms of deviation per unit)

Actually: PD = 1 cm, Centrad ≈ 0.5729 cm, Prism Degree = 1.75 cm

→ 1 Prism Degree is the largest displacement per unit

Constants of deviation:

- 1 Δ = 1 cm
- 1 ∇ (centrad) = 0.5729 cm = 1/100th of a radian (1 radian = 57.29°)

- $1^\circ d = 1.75 \text{ cm}$

Centrad derivation: 1 radian = 57.29°. A centrad is 1/100th of that arc. So 1 centrad = 0.5729 cm displacement.

Conversion Between Units

$$Q \times K = Q_n \times K_n$$

- Q = prism power in original unit
- K = constant of deviation for that unit
- Q_n = desired prism power
- K_n = constant of deviation for desired unit

Prentice Rule (Prismatic Effect from Decentration)

$$P = h \times D$$

- P = prismatic effect (in prism diopters)
- h = decentration (in cm)
- D = lens power (in diopters)

Base direction:

- Plus lens: base toward the direction you moved the optical center
- Minus lens: base opposite the direction you moved the optical center

Decentration	Plus lens	Minus lens
+ (same direction as prism)	same	opposite
- (opposite direction)	opposite	same

Worked Examples

1. Find prismatic effect:

-8.00 D lens, patient looks 12 mm below optical center:

$$P = h \times D = 1.2 \text{ cm} \times 8.00 = 9.6 \Delta \text{ base down}$$

2. Find decentration for desired prism:

+6.00 D lens, need 2 Δ base in:

$$h = P/D = 2/6 = 0.33 \text{ cm} = 3.3 \text{ mm}$$

3. Sphero-cylinder with decentration:

+3.00 S – 4.00 C × 90, 4 mm in, 2 mm up:

- Sphere component: H: $P = 0.4 \times 3.00 = 1.2 \Delta \text{ BI}$; V: $P = 0.2 \times 3.00 = 0.6 \Delta \text{ BU}$
- Cylinder (axis 90, power affects 180 meridian only): H: $P = 0.4 \times (-4.00) = -1.6 \Delta \text{ BO}$; V: 0
- Net: $H = 1.2 + (-1.6) = -0.4 \Delta \text{ BO}$; $V = 0.6 \Delta \text{ BU}$

Combining Prisms

Parallel prisms (same or opposite base direction):

- Same base direction → add powers
- Opposite base direction → subtract; base follows the larger power

Example: $3 \Delta \text{ BI} + 4 \Delta \text{ BO} = 1 \Delta \text{ BO}$

Non-parallel prisms (using Pythagorean theorem):

$$R = \sqrt{V^2 + H^2}$$

Example: $3 \Delta \text{ BI} + 4 \Delta \text{ BU}$ in front of OD:

$$R = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \Delta \text{ BU and I @ } 53.1^\circ$$

Optical Effects of a Prism

1. **Optical illusion** — object appears displaced toward the apex
2. **Prism aberration** — a circle viewed through a prism appears slightly oval; blurring along the base-apex line
3. **Metamorphopsia** — rotating a prism while observing distorts the object; appears to shrink or grow

28. Photometry

Photometry is the science of measuring light in terms of its perceived brightness to the human eye. **Radiometry** measures optical radiation in purely physical terms (watts).

Three Aspects of Photometry

Quantity	What it measures	Unit
Luminous flux	Rate of flow of light energy from a source	Lumen (lm)
Luminous intensity	Light emitted per unit solid angle	Candela (cd)
Illuminance	Light received per unit area of a surface	Foot-candle (fc) or Lux

Luminous Flux

$$F = 4\pi \times I$$

- **F** = total flux (lumens)
- **I** = luminous intensity
- **π** = 3.1416

Flux in practical terms: Lumens = (Radiation power in watts) $\times$ 683 lumens/watt $\times$ (luminous efficiency)

Efficacy = lumens/watt (typical incandescent: ~17 lm/W)

Solid Angle

$$\omega = A / r^2$$

- **A** = area on sphere surface
- **r** = radius of sphere

Unit: steradians (sr)

Luminous intensity = Flux / Solid angle = F/ω

Law of Inverse Square (Illuminance)

$$E = I / d^2$$

- **E** = illuminance (foot-candles or lux)
- **I** = luminous intensity (candela)
- **d** = distance from source to surface

Physical meaning: When you double the distance from a light source, illuminance drops to 1/4. Triple the distance → 1/9 the illuminance. This is why a near object looks so much brighter than a distant one.

Cross-subject connections:

1. This is the same inverse-square law used in low vision to counsel patients on task lighting — moving a lamp from 2 feet to 1 foot quadruples the light on the page
2. Illuminance affects visual acuity; low illuminance reduces contrast sensitivity

Cosine Law of Illumination

$$E = I / d^2 \times \cos \theta$$

Where **θ** = angle between the incident light and the normal to the surface.

When the surface is tilted relative to the incoming light, you receive less illumination (the same amount of light spreads over a larger effective area).

Law of Intensity (Photometer Principle)

The candle powers of two light sources are **directly proportional to the squares of their distances** from the photometer screen when they produce equal illumination:

$$I_1/I_2 = d_1^2/d_2^2$$

A **photometer** is an instrument for comparing the luminous intensities of two light sources. In the classic Bunsen photometer, a greasy-spot card is placed between two lamps; when both sides look equally bright, the proportion above holds.

Foot-Candle

- British unit of illuminance
- 1 foot-candle = 1 lumen per square foot = illumination of a surface 1 foot from a 1-candlepower source perpendicular to the surface

High-Yield Example

Two lamps: lamp A ($I = 40$ cd) at 2 m from photometer, lamp B at unknown distance, equal illumination. Find distance of lamp B if $I_B = 160$ cd:

$$I_A/I_B = d_A^2/d_B^2$$

$$40/160 = 4/d_B^2$$

$$d_B^2 = 16$$

$$d_B = 4 \text{ m}$$

29. Optical Instruments

1. Lens Clock (Lens Gauge)

Also called the **Geneva lens gauge**. Measures the surface curvature of a lens and converts it to power.

- Calibrated to index **$n = 1.53$** (crown glass)
- Three pointed probes: outer two fixed, center one moves
- Black scale = convex surfaces (plus power)
- Red scale = concave surfaces (minus power)
- When clock is rotated and hand does not move → **spherical surface**
- When clock is rotated and hand moves → **cylindrical surface**

Physically measures: sagittal height (depth) of the surface. Converts to power because sag relates to curvature, which relates to power.

True vs. Marked Power Formula (for non-standard index):

$$T/D = (n - 1) / 0.530$$

- **T** = true power of the surface

- **D** = marked power (gauge reading $F_1 + F_2$)
- **n** = actual index of refraction of the lens

When to use: When measuring a polycarbonate or high-index lens on a lens clock calibrated for crown glass (1.53). The clock reading will be wrong — you must correct it.

2. Lensometer (Focimeter)

Measures the **vertex power** (back vertex power) of a lens. Works on the Badel principle plus an astronomical telescope to detect parallel rays.

Lensometer Formula:

$$x = (f')^2 / x'$$

- **x** = distance from the mire to the anterior principal focus (APF) of the standard lens
- **x'** = distance from the posterior principal focus to the virtual image
- **f'** = focal length of the standard lens

Practical rule:

- Plus lenses: mires must be moved **toward** the standard lens
- Minus lenses: mires must be moved **away from** the standard lens

3. Lens Caliper

Measures the thickness difference between the center and edge of a lens. Reads in 1/10 mm (1 point reading).

Sagitta Formula:

$$\text{Sag} = (h^2 \times F) / (2000 \times (n - 1))$$

- **h** = half of the lens diameter
- **F** = lens power
- **n** = index of refraction

Sagitta is the depth of the "curve" carved into a flat surface to create the lens's curved shape.

To find lens power from caliper reading:

$$F = \text{Sag} \times 2000 \times (n - 1) / h^2$$

Where Sag = CT – ET (center thickness minus edge thickness)

4. Hand Neutralization (Trial Case Method)

Used to determine lens power by observing "with" or "against" movement when the lens is rocked in front of a target.

30. Transposition

Transposition means rewriting the same prescription in a different but optically equivalent form without changing what the patient needs.

Flat (Simple) Transposition

Three-step rule:

1. Add sphere and cylinder powers algebraically → new sphere
2. Change the sign of the cylinder
3. Rotate the axis by 90°

Example: $-0.50 \text{ S} - 1.50 \text{ C} \times 90$

1. $-0.50 + (-1.50) = \mathbf{-2.00 \text{ S}}$ (new sphere)
2. Change sign: $\mathbf{+1.50 \text{ C}}$
3. $90 + 90 = \mathbf{\times 180}$

→ Result: $\mathbf{-2.00 \text{ S} + 1.50 \text{ C} \times 180}$

Cross-Cylinder to Sphero-Cylinder

Write the two cylinder powers as sphere and cylinder:

- Take either cylinder as the sphere
- Subtract to find cylinder power, note axis of remaining one

Practice set from MCU 2024:

- $+3.75 \text{ C} \times 30 = +3.75 \text{ S} - 3.75 \text{ C} \times 120$ (or equivalent)
- $+1.75 \text{ C} \times 90 = +1.25 \text{ C} \times 180$ form → transpose first then combine

Single Vision Toric Transposition

Rule: The sign of the base curve must match the sign of the cylinder in the prescription. If it does not, flat-transpose first.

The base curve always carries the cylinder in the final toric form.

Formula structure:

$$\frac{F_{\text{toric surface}}}{B \pm \text{spherical surface}}$$

Bifocal Toric Transposition

Principle: The base curve must remain spherical (so it can carry the segment for the add power).

Two types:

1. **Fused bifocal** — segment in the **front** surface → base curve is **always positive (+)**
2. **Unfused bifocal (Ultex)** — segment in the **back** surface → base curve follows the back

Rules:

1. Sign of base curve must be **opposite** to sign of the cylinder power (flat-transpose if needed)
2. Base curve should be spherical — in front if plus, at back if negative

Ultex Bifocal Toric Transposition

$$\frac{F_{\text{toric surface}}}{B \text{ minus spherical surface}}$$

Steps:

1. Check if the prescription is in plus cylinder form
2. Copy the base curve below the line
3. Find a number that, when added to the base curve, gives the sphere component with the axis 90° away from the original
4. Add the two cylinders and copy the original axis

Reading Addition Conversion

Near Rx only: $D_{Rx} + ADD$

TRA (Total Reading Addition): Convert to near power by adding full ADD

Intermediate power: $\text{Sphere}(\text{distance}) + \frac{1}{2}(ADD)$

31. High-Yield Cheat Sheet

Formula Quick Reference

Formula	What it's for
$V = 1/d$	Vergence
$D = 1/f \text{ (m)}, 40/f \text{ (in)}, 100/f \text{ (cm)}, 1000/f \text{ (mm)}$	Diopter conversions
$1/f = 1/p + 1/q$	Thin lens and mirror
$M = -q/p$	Magnification
$n_1 \sin \theta_1 = n_2 \sin \theta_2$	Snell's law
$\sin \theta_c = n_2/n_1$	Critical angle
$\text{Images} = 360/a - 1$	Multiple images in plane mirrors
$\text{Deviation} = 180 - 2i$	Deviation by plane mirror
$\text{Dev} = a_A(n - 1)$	Prism deviation
$P = hD$	Prentice rule (prismatic effect)
$E = I/d^2$	Law of inverse square (illuminance)
$E = I/d^2 \cos \theta$	Cosine law of illumination
$I_1/I_2 = d_1^2/d_2^2$	Law of intensity (photometer)
$D_{eq} = F_1 + F_2 - (t/n)F_1F_2$	Equivalent power
$D_{BV} = F_1 + F_2 + (t/n)F_1^2$	Back vertex power
$D_{FV} = F_1 + F_2 + (t/n)F_2^2$	Front vertex power

$SE = S + \frac{1}{2}C$	Spherical equivalent
$y = kx$	Abney's formula (pinhole)
$\tan \theta = h/d$	Angular elevation
$Sag = h^2F / 2000(n-1)$	Sagitta (lens caliper)
$T/D = (n-1)/0.530$	True vs. marked power (lens clock)
$R = \sqrt{(V^2 + H^2)}$	Combined non-parallel prisms

Theories of Light (Memory Hook)

N-H-M-P (Newton → Huygens → Maxwell → Planck)

- Newton: **bullets** (corpuscles)
- Huygens: **waves** through ether
- Maxwell: **electromagnetic** waves
- Planck: **packets** (quanta/photons)

Shadow Rules Summary

- Point source → only umbra
- Extended source → umbra + penumbra
- Object closer to screen → smaller shadow
- Source closer to object → larger shadow

Eclipse Memory

- **SME** = Solar eclipse (Moon in middle between **S**un and **E**arth)
- **SEM** = Lunar eclipse (**E**arth in middle between **S**un and **M**oon)

Prism Constants

Unit	K (cm at 1 m)
------	---------------

Prism Diopter (Δ)	1.00 cm
Centrad (∇)	0.5729 cm
Prism Degree ($^{\circ}$ d)	1.75 cm

Index of Refraction Quick Table (Board Pearls)

- Air = 1.000
- Water = 1.333
- Crown glass = 1.523
- Polycarbonate = **1.586**
- Light flint = 1.58
- Dense flint = **1.65**
- Diamond = **2.417** (lowest critical angle $\rightarrow$ most internal reflection $\rightarrow$ most brilliance)
- Cornea = 1.376; Aqueous = 1.336; Vitreous = 1.336
- Lens cortex = 1.386; Nucleus = 1.406

Commonly Tested Numbers

- 360/60 – 1 = **5 images** (mirrors at 60 $^{\circ}$)
- Dense flint glass critical angle = **37.31 $^{\circ}$**
- Diamond critical angle = **24.4 $^{\circ}$**
- Lens clock calibrated to = **1.53**
- Abney's k in inches = **0.008**, in cm = **0.01275**
- 1 radian = **57.29 $^{\circ}$** (centrad base)
- Speed of light = **186,000 miles/sec** = **300,000 km/sec**

End of Theoretical Optics Consolidated Review

WAA | Board Review 2026 | Personal study notes